

Volleyball-Group-Activity-Recognition

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Abstract—A condensed summary of the Volleyball Group Activity Recognition project: the data pipeline, the two completed baselines (B1: single-frame ResNet-50; B3: person-then-group crop classifier), and the two silent data bugs whose discovery and repair took Baseline 3 from 23% to 61% test accuracy. The dataset (Ibrahim et al., CVPR 2016) contains 55 videos / 4,830 clips with 8 scene-level group activities and 9 player-level actions.

Keywords—Volleyball, Group Activity Recognition, CNN, RNN, LSTM, ResNet50, Classification, Multiclass Classification, Transfer Learning

1. Problem & Dataset

Volleyball Group Activity Recognition is a hierarchical classification problem: predict the scene-level group activity (8 classes: 1/r-pass, -spike, -set, -winpoint) and the per-player action (9 classes: blocking, digging, falling, jumping, moving, setting, spiking, standing, waiting). The dataset comes from “A Hierarchical Deep Temporal Model for Group Activity Recognition” (Ibrahim et al., CVPR 2016; [official repo](#), [Kaggle mirror](#)).

Key facts: **55 videos, 4,830 clips** (train/val/test = 2,152/1,341/1,337 clips over disjoint video splits of 24/15/16). Each clip has **41 frames**; the clip folder is named after its annotated *middle frame*. Both label levels are heavily skewed: standing is $\approx 68\%$ of person actions, and 1/r_winpoint are $\approx 2.5\times$ rarer than the other group classes. The group classes come in **left/right mirror pairs**, so horizontal geometry is part of the label — no horizontal flips, no crops that discard court width.

2. Data Pipeline

Three raw annotation sources (action detections, person detections, player tracking) are unified by a two-stage parser (`src/json_parser.py`) into one master JSON keyed `video_id/clip_id`, then enriched with the group-activity label per clip and cached as a pickle (~ 247 MB) for instant metadata loading. Frames are served either from a memory-mapped LMDB (local) or straight from disk (Kaggle); both backends expose the identical `VolleyballDataset`, which covers every baseline through flags: full frames or per-player crops, 1 or 9 frames per clip. A custom collate pads the variable player count per clip and returns a validity mask so padded slots never reach the loss or the pooling.

3. Milestone: Two Silent Data Bugs

Both baselines were initially stuck near chance level regardless of hyperparameters. Two input-corruption bugs — neither of which ever raised an error — were found and fixed in July 2026:

- Label corruption (62% of the dataset).** The scene-label merge matched clips against a *flat* frame-name dictionary, but frame names are only unique within a video; 3,001 of 4,830 clips received another clip’s label. Fixed by keying the lookup per video and matching each clip’s own middle frame; all 4,830 labels now verify against the per-video ground truth.
- Transforms destroyed the input.** Person crops (median 56×127 px) went through a short-side resize + 224 center crop that kept only $\approx 40\%$ of the player’s height — classifying arm/leg actions from torsos. Full frames similarly lost over half the court width, deleting the left/right signal. Both pipelines now warp directly to 224×224 with mild photometric augmentation only.

4. Baseline Roadmap

Table 1 lists the planned progression of models, each adding temporal and/or player-level structure on top of the previous one.

Table 1. Baseline models. B1 and B3 are complete; B4–B8 pending.

Model	Temporal	Player	Scene
B1	—	—	Image clf.
B3	—	Crop clf.	Concat-pool + MLP
B4	LSTM	—	LSTM
B5	LSTM/player	Max-pool	NN
B6	LSTM/image	Max-pool	LSTM
B7	$LSTM_1 + LSTM_2$	Max-pool	$LSTM_2$
B8	$LSTM_1 + LSTM_2$	Team-split	$LSTM_2$

5. Baseline 1: Single-Frame Classifier

Fine-tunes a pretrained ResNet-50 on the clip’s middle frame for the 8 group classes. Two-stage schedule: linear probe (head only, 10 epochs, $lr\ 10^{-3}$), then partial fine-tune of layer3/4 + head (up to 100 epochs, backbone $lr\ 2.5 \times 10^{-4}$, head $3\times$, cosine annealing, batch 16, frozen BatchNorm held in eval).

After the label fix but still on the legacy transforms, B1 reached **62.98% test accuracy / 0.641 macro F1** (up from 20.8% / 0.196 under corrupted labels); Fig. 1 shows its confusion matrix. A retrain on the fully corrected pipeline (full-court warp, batch 16, rescaled lr) is in progress and will replace these numbers.

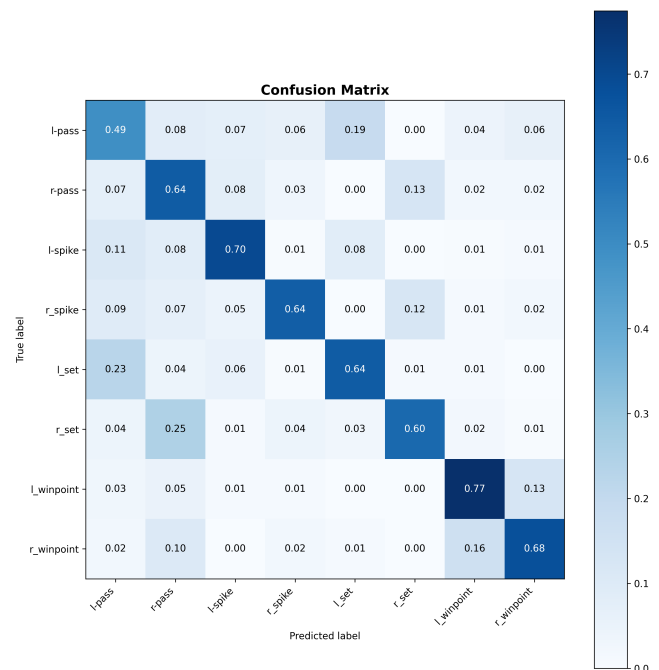


Figure 1. Baseline 1 test confusion matrix (corrected labels, legacy transforms).

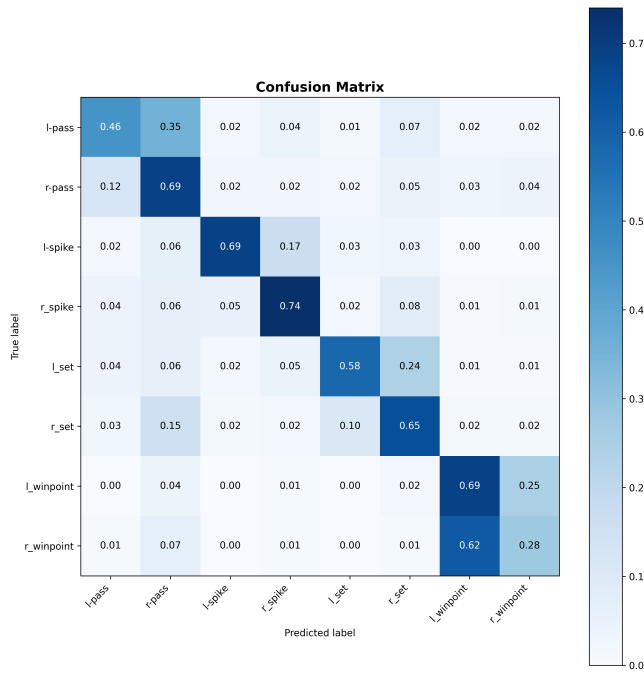
6. Baseline 3: Person-Then-Group

Stage A fine-tunes ResNet-50 on individual player crops for the 9 person actions with inverse-frequency class-weighted CE (without it the loss collapses onto *standing*). **Stage B** freezes that backbone, extracts a $(P, 2048)$ feature matrix per clip, concatenates masked max- and mean-pooling across players (2×2048), and trains an MLP ($4096 \rightarrow 512 \rightarrow 8$, Dropout 0.4) with class-weighted CE for the group classes. SGD (momentum 0.9, Nesterov), 50 epochs and patience 10 per stage, batch 16 clips, nn.DataParallel on multi-GPU hosts.

Table 2. Baseline 3 before/after the data-pipeline fixes.

	run 1 (pre-fix)	run 2 (fixed)
Stage A val acc / F1	59.7% / 0.387	70.4% / 0.512
Stage B val acc / F1	39.2% / 0.370	60.0% / 0.584
Test acc / F1	23.1% / 0.227	60.7% / 0.589

Stage A’s gain isolates the transform fix (its labels were never corrupted); Stage B compounds the corrected labels with the stronger backbone. Fig. 2 shows the dominant remaining error mode: left/right mirror confusion within each activity pair, worst on the rare `r_winpoint`. The other open issue is **overfitting** (train $\approx 95\%$ vs. val 58–70%) — a regularization problem for a follow-up run, not a data issue.

**Figure 2.** Baseline 3 test confusion matrix (run 2, fixed pipeline).

7. Next Steps

(i) Finish the B1 retrain on the corrected pipeline; (ii) attack B3’s overfitting (stronger augmentation, higher dropout, earlier stopping); (iii) proceed to the temporal baselines B4–B8 — noting that player tracking covers only $\approx$ half of each clip’s frames, so the detection-fallback path of the loader must be label-corrected before B5.